

New Regulations Concerning Airborne Particle Counters (ISO 21501 and ISO 14644)



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Table of Contents

- **Introduction**
- **Whitepaper:** ISO 21501 – A Standard Methodology to Optical Particle Counter Calibration and What it Means to Cleanroom Owners
- **Whitepaper:** Optical Particle Counters and Counting Efficiency
- **Whitepaper:** ISO 21501-4 vs ASTM F 328-98 (2003)

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Introduction

In May, 2007, the International Standards Organization (ISO) introduced a new standard for the calibration of air and liquid particle counters. This new standard, ISO 21501, incorporates a number of new tests that are designed to reduce count variability between different instruments. Specifically, the standard specifies that counting resolution and counting efficiency tests be performed at routine calibration cycles, typically every six months. In 2011, ISO21501 compliant calibrations for air particle counters will become mandatory for aseptic filling under EU-GMP Annex 1 and FDA cGMP Guidance.

In preparation for this new standard, in late 2005 when the ISO 21501 standard was still in draft, Hach Company incorporated the standard's performance criteria within the specifications for all air particle counter development projects. In addition, a technical audit was conducted of the entire product portfolio and where practical, existing product designs were enhanced to comply with the new standard.

Hach, as the global leader in particle counting, has the largest installed base of air particle counters in the world. To support this install base, Hach's new calibration system, CoreCal, was designed and developed to perform full ISO 21501 compliant calibrations. The CoreCal system, backed by globally deployed calibration tools, procedures and technician training, has enabled all of Hach's worldwide service teams to provide ISO-21501 compliant calibrations in the field.

The purpose for this resource guide is to help you, our customer, better understand ISO 21501 and its potential impact on air particle counting to help you "Be Prepared"



ISO 21501 – A Standard Methodology to Optical Particle Counter Calibration and What It Means to Cleanroom Owners

Tony Harrison and Bob Latimer

INTRODUCTION

ISO 21501 is a new family of standards describing the instruments and calibration requirements for determining particle size distribution using light interaction methods. It represents the culmination of work by instrumentation manufacturers and industry users and comes at a critical time for the life science industry with the increasing trend for real-time air particle monitoring in cleanrooms using light scattering air particle counters.

Air Particle Counters and ISO 21501

In comparison to liquid particle counters, the calibration of air particle counters presents greater challenges due to the need to generate air samples containing sub-microscopic particles of homogenous size and distribution. Although the technology of air particle counting is well understood, the ability to calibrate any two air particle counters so that they produce the same results when sampling the same air sample has proven to be challenging, bringing into question the accuracy of these instruments. ISO 21501 now delivers a calibration method that can significantly improve the repeatability and reproducibility of air particle counters.

Liquid Particle Counters and ISO 21501

ISO 21501 also applies to liquid particle counters used for determination of particulate contamination in infusions and injections. Until recently, the calibration requirements [known as "IST" methods] for liquid particle counters used to test infusions and injections were described in detail in the United States Pharmacopoeia (USP) chapter <788>. However, in the interest of international harmonization of the pharmacopoeias, the details of these IST calibration methods have been removed in order to simplify the text of USP <788>. ISO 21501 now offers an alternative to these IST tests and establishes calibration methods to ensure accurate and repeatable performance of liquid particle counters.

BACKGROUND

Optical instrumentation has been used to determine particle contamination in air and liquids in the life science industry for many years. In addition, the correlation between airborne particles and final product quality has long been recognized in the semiconductor, flat panel display and hard disk storage manufacturing industries, where improvement of air quality (reduction of particulate contamination) has led to increases in final product yield.

Differing techniques are used to determine the number and size of particles depending on the size of particles that are of interest (see Figure 1).

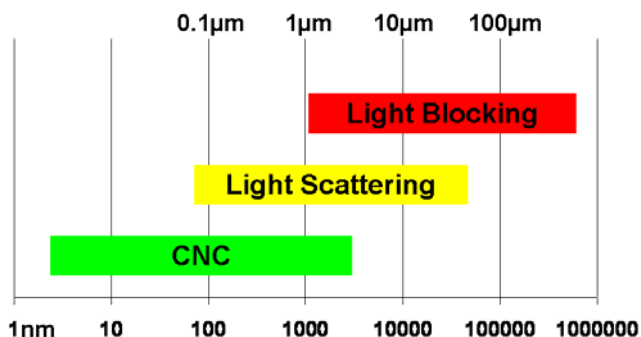


Figure 1 - Particle size range and counting techniques

In liquid particle counting for infusions and injections, the sizes of interest are $\geq 10\mu\text{m}$ and $\geq 25\mu\text{m}$, whereas for the life science industry, the sizes of interest for cleanroom air particle cleanliness are $\geq 0.5\mu\text{m}$ and $\geq 5\mu\text{m}$. Higher sensitivities are required for semiconductor manufacturing plants where cleanroom and

mini-environment air is routinely monitored at 0.1 µm and lower. Hard disk manufacturers typically monitor to around 0.2 µm to 0.3 µm and flat panel display manufacturing environments monitor to 0.3 µm and 1.0 µm.

USP <788>, EU 2.9.19 and JP 6.07 recognize that light obscuration is suitable for liquid particle counting in infusions and injections, whereas ISO 14644-1 recognizes that light scattering particle counters are appropriate for determining airborne contamination in cleanrooms.

There is a requirement to follow the guidelines in EU GMP and cGMP for cleanroom users that aseptically manufacture pharmaceutical products for the European and American markets. Both documents define the airborne particulate count limits for different cleanroom operations, but neither defines the methods required to determine these count limits, nor do they define the instrument to be used and how it should be calibrated. However, EU GMP states that ISO 14644-1 should be used for methodology to determine cleanroom air particle cleanliness classification and that ISO 14644-2 should be used for methodology for demonstrating continued compliance. The introduction in ISO 21501-4 states, *"Monitoring particle contamination levels is required in various fields, e.g. in the electronic industry, in the pharmaceutical industry, in the manufacturing of precision machines and in medical operations. Particle counters are useful instruments for monitoring particle contamination in air. The purpose of this part of ISO 21501 is to provide a calibration procedure and verification method for particle counters, so as to minimize the inaccuracy in the measurement result by a counter, as well as the differences in the results measured by different instruments."* The scope of ISO 21501-4 states, *"Instruments that conform to this part of ISO 21501 are used for the classification of air cleanliness in cleanrooms and associated controlled environments in accordance with ISO 14644-1"*. So the importance of ISO 21501 to cleanroom users looking to follow the guidance in GMP is evident.

Equally the scope of ISO 21501-2 states, *"Instruments that meet this standard are used for the evaluation of cleanliness of pharmaceutical products (injections, water for injections, infusions), as well as the measurements of number/size distribution of particles in various liquids."* So the importance of ISO 21501 to those in the pharmaceutical industry manufacturing injections, water for injections or infusions is also evident.

WHAT STANDARDS EXIST? WHAT IS ISO 21501 REPLACING?

ISO 14644 is a widely used standard for cleanroom classification using optical particle counters. Despite the existence of ISO 14644, prior to the ratification and introduction of ISO 21501 at the beginning of 2007, there were no ISO standards dealing with calibration and performance of the optical particle counters (OPC) used to classify cleanrooms to ISO 14644. Comprehensive non-ISO standards and calibration methods guidelines did exist however and have been employed by most major particle counter manufacturers. In summary, these standards are:

- ASTM F 328-98(2003) *"Standard Practice for Calibration of an Airborne Particle Counter Using Monodisperse Spherical Particles"* (withdrawn May 2007).
- IEST-RP-CC014.1 *"Calibration and Characterization of Optical Airborne Particle Counters"* (providing actual methods to perform the calibration).
- JIS B 9921:1997 *"Light scattering automatic particle counter"*, a Japanese standard which comprehensively deals with OPC design performance, most notably in the area of counting efficiency.

The counting efficiency parameter has presented the most significant variable when it came to the actual count accuracy of individual OPC's, especially air particle counters.

Counting Efficiency

OPC's typically feature a number of size channels into which particle counts are binned, each channel being calibrated to count particles greater than a specific particle size. Particle sizes are typically expressed in micrometers (µm). The term *counting efficiency* primarily refers to the ability of the OPC instrument to count particles at a specified size. Typically, calibration involves passing a continuous stream of standard, mono-

sized particles through the OPC's sensor, which results in a stream of electrical pulses, each pulse being proportional to the size of each particle. The mono-sized standard particles produce a distribution of pulse heights, the median of which is typically regarded as the appropriate channel calibration threshold for that size. Therefore, in the real world a particle exactly the same size as a given channel would have a 50% probability of being counted (see Figure 2a). As a result, OPC's calibrated in this manner are said to have a counting efficiency of 50%. Note however that this does not mean that the OPC will only count half of the particles in the real world.

ISO 21501 makes use of the specification for counting efficiency accepted in the JIS B 9921 standard. This states that the counting efficiency should be 50% $\pm$ 20% (i.e. between 30% $\rightarrow$ 70%) in the first channel (Figure 2a). Additionally, particles of between 1.5 X to 2.0 X the channel 1 particle size should be counted with an efficiency of 100% $\pm$ 10% (i.e. between 90% $\rightarrow$ 110%) in the first channel (Figure 2b.)

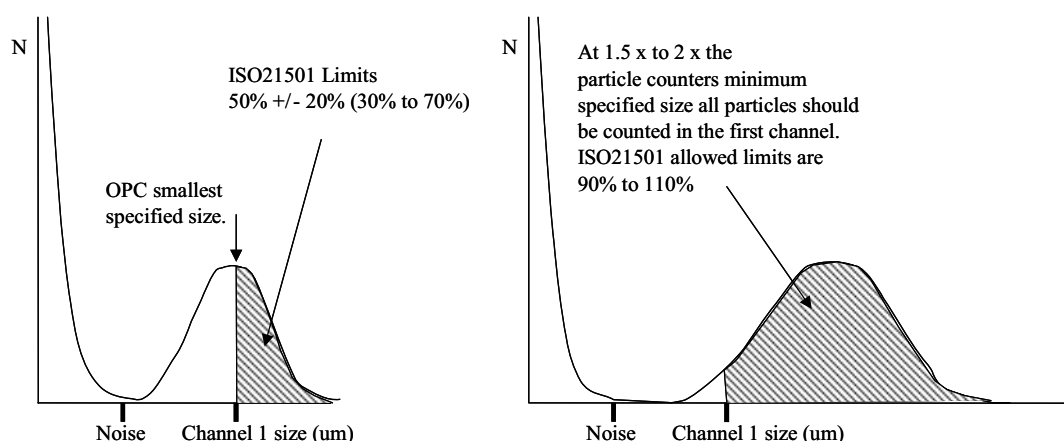


Figure 2a: The 50% calibration point 2b: Verifying 100% efficiency at a higher size

WHY WAS A NEW STANDARD REQUIRED?

Prior to ISO 21501, it was not required that counting efficiency be checked at each calibration interval. There are many things that can impact counting efficiency during the lifetime of an OPC; for example a slight opto-mechanical misalignment of the illumination source can go undetected. Therefore, the situation exists where even though a given OPC may correctly size particles, it may be undercounting - in effect missing some of the particles. The new ISO 21501 standard requires (among other elements) that the all-critical counting efficiency element be checked during calibration. To check counting efficiency, it is necessary that once calibrated for sizing characteristics using traceable size standards, the OPC under test must be run and compared to either an Electrostatic Classifier or an OPC instrument with higher sensitivity than the OPC under test. This OPC is considered to be a "secondary standard", having been formally compared to an Electrostatic Classifier and verified as having 100% counting efficiency at the size of interest, i.e. the channel 1 size of the OPC to be certified.

The full list of elements that ISO 21501 requires to be tested in addition to the basic size calibration are as follows:

- Counting efficiency
- Sizing resolution
- False count rate
- Concentration limit
- Sampling flow rate
- Sampling time
- Sampling volume

WHAT IS ISO 21501 AND WHAT IMPROVEMENTS WILL IT BRING?

To quote from the ISO 21501 standard:

“The purpose of this part of ISO 21501 is to provide a calibration procedure and verification method for particle counters, so as to minimize the inaccuracy in the measurement result by a counter, as well as the differences in the results measured by different instruments.”

Simply put, the ISO 21501 standard will ensure that OPC instruments will size and count particles correctly, using a traceable reference instrument. Different OPC models from different manufacturers will therefore closely correlate in terms of actual particle counts recorded. This presents a significant step forward in providing traceable, accurate OPC tools to classify and validate cleanrooms to ISO 14644.

WHO SHOULD ADOPT ISO 21501?

Manufacturers of products requiring processing or assembly all goods and materials within a cleanroom environment classified under ISO 14644 should require that OPC instruments be calibrated to the ISO 21501 standard. This is particularly applicable to the pharmaceutical manufacturing facilities employing sterile processing or filling lines.

Users of cleanrooms OPC's with questions or concerns regarding the transition to ISO 21501 should contact their particle counter supplier or cleanroom certifier.

ADDITIONAL INFORMATION

The ISO 21501 family of standards extends beyond air particle counters to include both scattering and extinction type liquid particle counters. The standard is split into four parts and all are available from ISO at <http://www.iso.org>.

ISO 21501 Determination of particle size distribution – Single particle light interaction methods -

- Part 2: Light scattering liquidborne particle counter
- Part 3: Light extinction liquidborne particle counter
- Part 4: Light scattering airborne particle counter for clean spaces

Hach Ultra manufactures a range of ISO 21501 compliant particle counters and is currently in the process of deploying an ISO 21501 field service capability for the calibration of existing products.

Particle counter owners and users with specific questions or concerns regarding ISO 21501 are invited to email the Hach Ultra ISO 21501 support team at iso21501@hachultra.com. Through this email address, one can access a panel of experts regarding ISO 21501 and receive prompt and accurate answers to questions.

CONTACT

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Optical Particle Counters and Counting Efficiency

Bob Latimer

INTRODUCTION

Typically, the back page of any analytical instrument's data sheet reveals a bewildering array of specifications. Optical particle counters (OPC) are no different. Many of the parameters used to define OPC performance such as flow rate are relatively easy to understand. Others such as coincidence loss are far less intuitive. However, the one parameter that stands alone as the champion of confusion is *counting efficiency*. The purpose of this article is to explain OPC counting efficiency in terms that are easy for all to understand.

COUNTING EFFICIENCY DEFINED

To understand OPC counting efficiency it is first necessary to examine the basic principle behind OPC calibration, which involves passing a continuous stream of standard, mono-sized, standard particles through the OPC sensor. This results in a stream of electrical pulses, each pulse being proportional to the size of each particle. The mono-sized standard particles produce a distribution of pulse heights, the median of which is typically regarded as the appropriate channel calibration threshold for that size. This calibration method is required by JIS B9921 and more recently by ISO 21501-4.^{1,2}

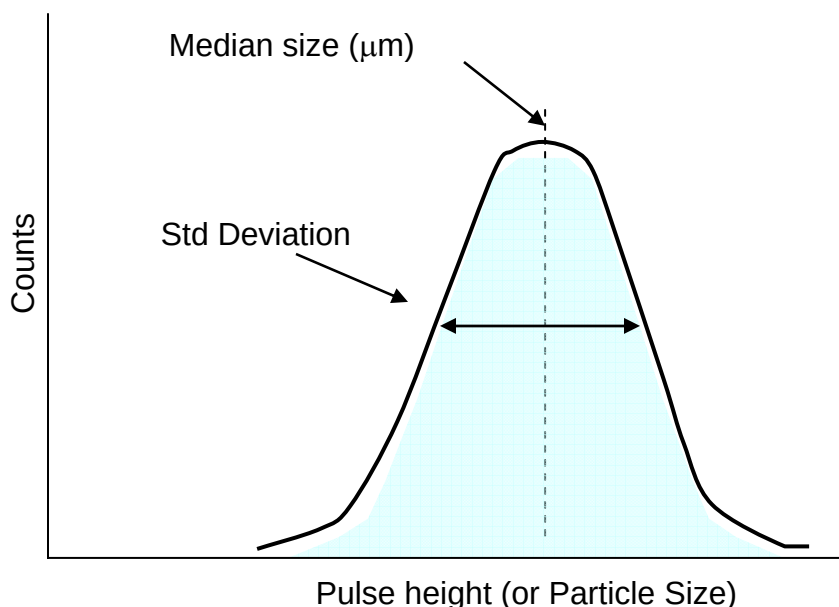


Figure 1: Mono-sized, standard particle distribution and the median calibration threshold

By calibrating an OPC in this manner, a single particle having exactly the same size as a given channel's counting threshold, would have a 50% probability of being counted in that channel as it passed through the OPC sensor. As a result, OPCs calibrated this way are by definition said to have a counting efficiency of 50%. However, for all particles even slightly greater than a given channel threshold, the probability of being counted in that channel quickly rises to 100%.

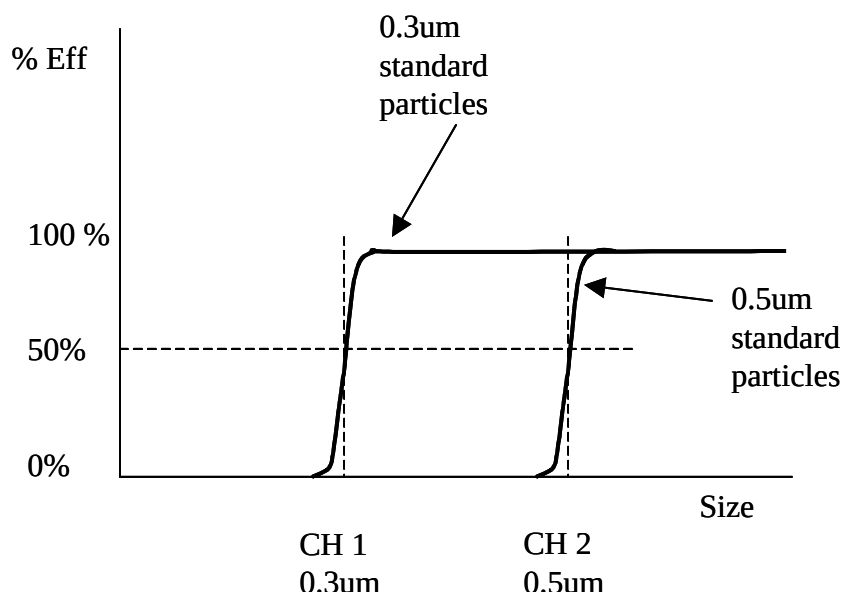


Figure 2: Particles greater than the size channel threshold are all counted with 100% efficiency

In practical terms for example, when examining the data sheet of an ISO 21501-4 compliant OPC with a minimum sensitivity of 0.3 μm , one would expect to see the parameter of counting efficiency specified similar to the following:

Counting Efficiency: 50% $\pm$ 20 % at 0.3 μm , (100% $\pm$ 10% at 1.5X minimum sensitivity)

Unfortunately, without understanding counting efficiency terminology, this may lead the reader to believe that the OPC will only count half of the 0.3 μm particles in a real sample. This is not true! In fact, all particles greater than 0.3 μm will be counted. Furthermore, to add to the confusion, the reader may also believe that the next channel (0.5 μm in this example) has a counting efficiency of 100% since 0.5 μm is greater than 1.5 times the minimum sensitivity. Again, this is not true.

The fact is, ALL channels in a properly calibrated OPC will, by definition, have a counting efficiency of 50%. However, this results in detection of particles greater than each channel threshold to be 100% for that channel (see Figure 2).

COUNTING EFFICIENCY AND PARTICLE MONITORING

It was stated above that OPC calibration involves passing a continuous stream of standard, mono-sized particles through the OPCs sensor. Figure 3 shows the OPC first channel set to the mean of such a 0.3 μm distribution. This is typically referred to as the 50% point and is the basis of the statement that the counting efficiency equals 50%.

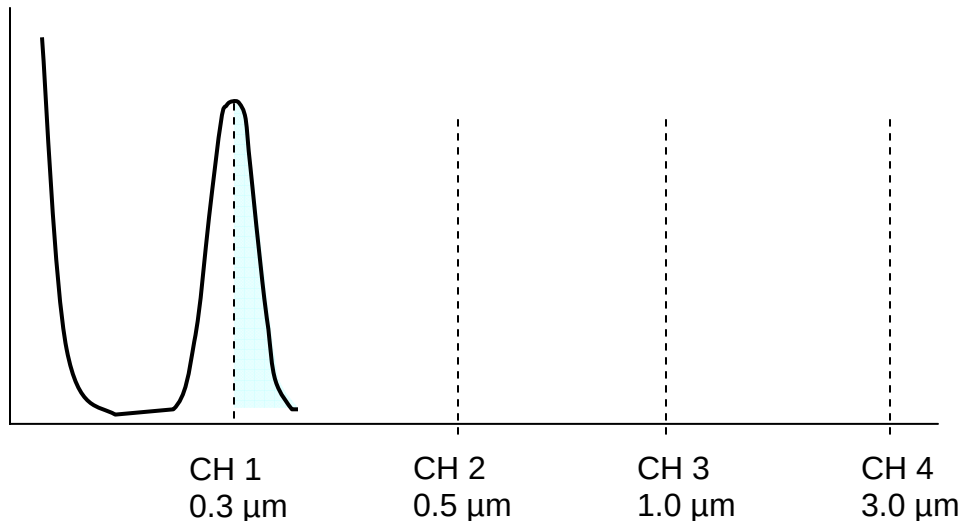


Figure 3: Threshold properly set for an OPC with a minimum channel size of 0.3 μm

In Figure 4, one can see that the same OPC, when challenged with 0.5 μm standard particles will also properly exhibit a counting efficiency equal to 50% for the 0.5 μm channel. Furthermore, because the distribution of particles in an ISO21501-4 compliant counter is narrow, it is easy to differentiate between 0.3 μm and 0.5 μm particles. As a result, ALL of the 0.5 μm standard particles are greater than the 0.3 μm threshold and it can therefore be stated that 0.5 μm particles will be counted with 100% efficiency in the first channel (the 0.3 μm channel), not the 0.5 μm channel.

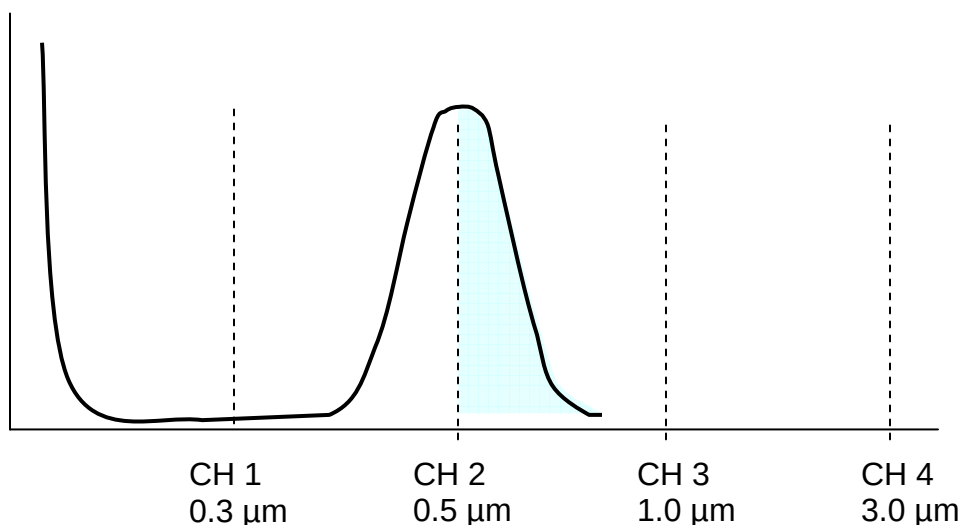


Figure 4. All channel thresholds set to the mean of standard sized particles per ISO21501-4 exhibit 50% counting efficiency regardless of the minimum channel size

Particles in air typically exhibit a size distribution such that the greater the size of the particles the fewer the number of particles are observed for a given volume of air. Figure 5 illustrates how the first channel of an ISO21501-4 calibrated OPC counts 100% of all particles greater than 0.3 μm and similarly Figure 6 shows that the second channel (0.5 μm) counts 100% of all particles optically greater than 0.5 μm . In other words, nothing is missed!

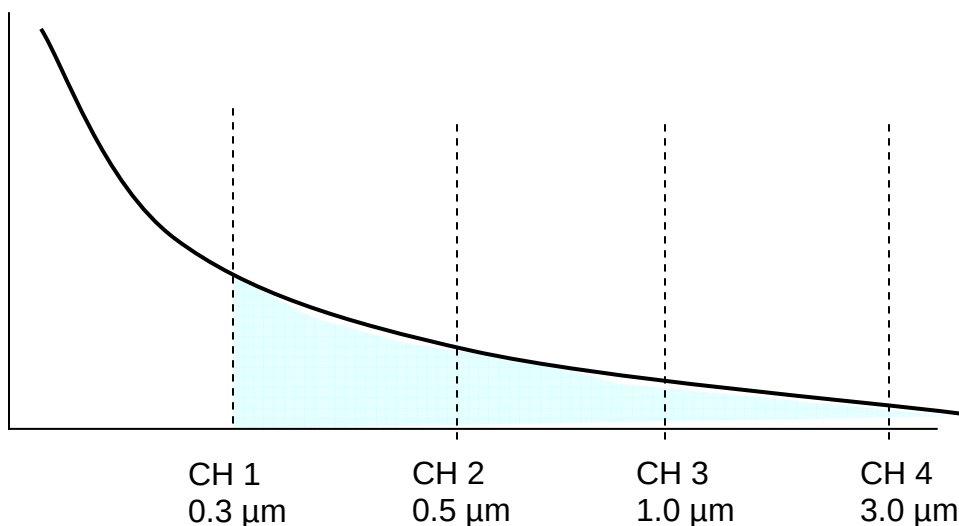


Figure 5 All particles greater than 0.3 μm are counted in the first channel

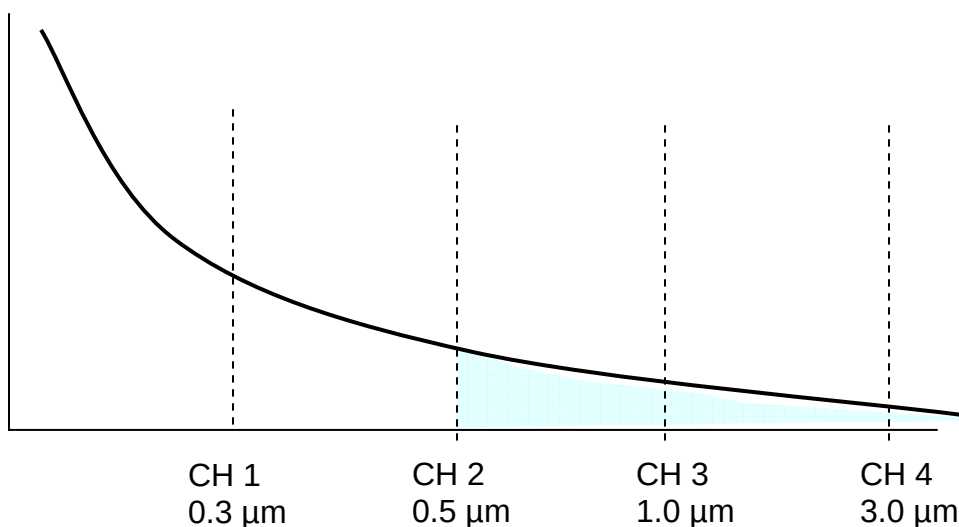


Figure 6: All particles greater than 0.5 μm are counted in the second channel

Regardless of the OPC's minimum specified size, all particles larger than the threshold are counted. Therefore, a particle counter specified with a first channel size of 0.5 μm will count 0.5 μm and larger particles with the same performance as a particle counter with a minimum specified size of 0.3 μm will count 0.5 μm . The specification for counting efficiency performance is independent of the minimum channel size.

CONCLUSION

This paper has demonstrated that an OPC calibrated using standard particles per ISO 21501 or JIS B9921 will exhibit 50% counting efficiency at ALL channels regardless of the minimum channel size.

Furthermore, in air samples taken from real world situations, such as controlled environments or cleanrooms, all channels in an OPC (calibrated to the aforementioned standards) will count 100% of the particles that are greater than the channel's threshold size regardless of the particle counter's minimum channel size

Thus, by definition, a properly calibrated particle counter with a minimum specified size of 0.5 μm will count 0.5 μm and larger particles with the same efficiency and performance as a particle counter with a smaller minimum channel size such as 0.3 μm . It is important to note that there is no advantage to over-specifying the minimum size threshold in an optical particle counter.

REFERENCES

- ¹ JIS B 9921, Japanese Standards Association, 1997
- ² ISO 21501-4, Light scattering airborne particle counter for clean spaces, 2007

ABOUT THE AUTHOR

Bob Latimer is the product manager for Air OPC with Hach Company's Particle Counting Business Unit with a focus on contamination monitoring for the Pharmaceutical, Semiconductor, Flat Panel Display and Hard Disk Storage industries.

An Electrical and Electronic Engineer by background, Bob has over 20 years experience in particle counting. After several years based in Taiwan working with Asia's electronic manufacturing industry, he is currently located at the business unit's headquarters in Grants Pass, Oregon.

Bob can be contacted via email at blatimer@hach.com





**ISO 21501-4 Determination of Particle Size Distribution
Single Particle Light-interaction Methods – Part 4
Vs.
ASTM F 328-98(2003) “*Standard Practice for Calibration of an Airborne Particle Counter
Using Monodisperse Spherical Particles*” (Withdrawn May 2007)**

Jarret R. Young

INTRODUCTION

The withdrawal of ASTM F 328-98 (Reapproved 2003) *Standard for Calibration of an Airborne Particle Counter Using Monodisperse Spherical Particles*, caused a void in the particle counting calibration methodologies. To answer this void ISO created a standard practice identified as ISO 21501-4. This document will contrast the differences in the processes.

ELEMENT COMPARISON

ASTM 328 Processes		ISO 21501-4
4.1 Sample Flow Rate	→	4.7 Sampling Flow Rate
4.2 Particle Sizing Accuracy	→	4.1 Size Calibration
	→	4.2 Verification of Size Setting
4.3 Particle Sizing Resolution	→	4.4 Size Resolution
4.4 Zero Count Rate	→	4.5 False Count Rate (Zero Count)
4.5 Particle Counting Efficiency (100%)	→	4.3 Counting efficiency (100 and 50%)
4.6 Particle Concentration Limit	→	4.6 Maximum Concentration
		4.8 Sampling Time
		4.9 Response Time
		4.10 Calibration Interval
		4.11 Test Report

Above you will see the basic differences of the two standards including the differences in the number of tests and the terminology. For example, ASTM uses the term zero count where ISO used false count. The arrows provide the corresponding element of ISO 21501 for direct comparison to the ASTM standard.

ELEMENT COMPARISON RATIONALE

ASTM 328 - 4.1 Sample Flow Rate

The Sample Flow Rate is the same process as 4.7 of the ISO standard. In both standards an external flow meter is used to measure the volumetric flow rate of the sample being taken. Hach Ultra utilizes the volumetric flow meter to accurately measure and calibrate the flow meter of the instrument under test. Volumetric flow measures allow for a more accurate measurement by removing the variance that altitude would cause.

ASTM 328 - 4.2 Particle Sizing Accuracy

4.1 and 4.2 of the ISO standard take into account not only the sizing accuracy of the instruments sensor but also the calibration point.

ASTM 328 - 4.3 Particle Sizing Resolution

4.4 of the ISO standard details the resolution and provides for specific values unlike the ASTM standard. Although both standards contain resolution as an element for calibration the nomenclature contained in the ISO standard is much more detailed and also is mandated as part of the result reporting.

ASTM 328 - 4.4 Zero Count Rate

This corresponds with 4.5 False Count Rate as listed by the ISO standard. In this section the False Count Rate is better defined and allows a “Confidence Level” to be assigned. Using a poison distribution the 95% Confidence Level is measured and presented to the end user. This process closely follows the process of JIS 9921B.

ASTM 328 - 4.5 Particle Counting Efficiency

This standard corresponds with 4.3 Counting Efficiency at 100% and 50% for ISO 21501. ASTM required only a 100% Counting Efficiency test. In the ASTM test the purpose, as in the ISO standard, was to ensure an accurate measure of the stream value. The test validated that the sensor system was sampling the full stream of the sample. Historically, manufacturers and calibration laboratories called this a “Total Count” test. In the ISO standard the 50% efficiency is also measured to ensure that the electronics associated with the system are also able to define the 50% point accurately. This is part of a redundancy where the resolution testing is also part of this definition. The efficiency testing verifies the accuracy of the sensor and its supporting system to define and count a particle in the stream.

ASTM 328 - 4.6 Particle Concentration Limit

A sensor system may change through degradation of the optic path as a result of either damage or contamination through usage. Maximum concentration testing for ISO requires testing at every interval of calibration rather than as a design test only.

The additional testing requirements of the ISO standard adds reliability by requiring that the original design test requirement be performed annually. Elements 4.8 to 4.11 are discussed below.

ISO 21501-4 ADDITIONS

ISO 21501-4 - 4.8 Sampling Time

Sampling Time is measured by simply comparing the sample time as directed by the instrument under test to a calibrated stopwatch or timing device. This test would not be performed on an instrument that does not have a timing circuit, such as a remote optical particle counter (OPC). Since the flow volume in most airborne counters is controlled by a set flow rate and a period in which the counter module is active, timing is a crucial component of the resulting sample volume.

ISO 21501-4 - 4.9 Response Time

Response Time is a measure of the clean-up rate of a sensor and its ability to zero count in a timely manner after exposure to a high level of contamination. This is the recovery time of the system if exposed to high level of counts. This test can be destructive to the sensor. Exposure levels of the testing are well above application levels. This does however; provide information on the sensor's ability to clean-up. More specifically, in the airborne industry it helps at the design phase when looking at the mechanical design. This design level test would provide information to the designer to locate the existence of particle traps. Particle traps in the system would allow particle contaminants to collect until they broke loose and were counted in error. In some cleanroom applications where the count tolerance is low, the added counts of 1 or 2 could make the difference in an acceptable batch.

Under the current parameters of the ISO testing, it is considered a destructive test and therefore Hach Ultra has retained this test as a design test only.

ISO 21501-4 - 4.10 Calibration Interval

This is the first time that a calibration standard added a suggestion in this format. ISO 21501-4 states; "4.10 Calibration interval, it is recommended that the calibration interval of an LSAPV be one year or less." Previously, the standards have suggested to the end users that they use the manufactures recommendation. This is a great addition as it now places a format to the information. Most of the OPC manufactures have been providing this information in the past based on customer feedback.

ISO 21501-4 - 4.11 Test Report

Where ISO 17025 has 11 points, ISO 21501 has only 7. The ISO 17025 standard governs the laboratory as a whole and mimics ISO 9000 series of quality management system (QMS) requirements.

CONCLUSION

The two standards, ASTM 328-98 and ISO 21501 are very similar in both methodology and requirements. ISO 21501 adds more detail resulting in increased data consistency. Both ASTM and ISO provide methods for the calibration and monitoring. Both standards are useful in ensuring that applicable industries are consistent in their measures, methods and reporting. As a manufacturer or calibration lab, the ISO standard's elements are used in whole or in part to provide customers the flexibility to meet their specific compliance needs.